Faculty: Computers and Informatics Class Year: 2024/2025 Subject Name: Computer Networks Department: Information Technology Course Code: IT202	Midterm Exam	Date: 15 Apr. 2025 Time Allowed: 60 min. No. of Pages: 2 No. of Questions: 40 Full Mark: 15 Points
Student Name:		Grade:
Section:		

Please answer all questions by selecting the correct answer:

1. How long does it take a packet of 50 MB to propagate over a link from point A to another point B? The distance between the two points equals to 60 km, propagation speed = 3.5x10^4 m/s, and transmission rate = 4 Mbps.

Answer: ..... Seconds.

2. In question 1, does this delay depend on packet length?

(a) Yes (b) No

3. In question 1, does this delay depend on transmission rate?

(a) Yes (b) No

4. In question 1, what is the transmission delay?

Answer: ..... Seconds.

5. In question 1, if the processing delay = 0.35 seconds and queuing delay = 0.97 seconds, what is the total transmission delay?

Answer: ..... Seconds.

Two hosts are connected via a router to send a 4-KB packet from one host to the other. The router has two links and each link is connected with one host. The first link is 300 meters long and the second link is 140 meters. The first link has a transfer rate of 200 Kbps, while the second link has a transfer rate of 100 Kbps. The router's queuing delay is 7 ms and the propagation speed for each link is 25,000 m/s. Use these details to answer questions (6 - 10).

6. What is the propagation delay for the second link?

Answer: ..... Seconds.

7. What is the total propagation delay?

Answer: ..... Seconds.

8. What is the transmission time for the first link?

Answer: ..... Seconds.

9. What is the nodal delay for the second link?

Answer: ..... Seconds.

10. What is the total end to end transfer delay?

Answer: ..... Seconds.

11. What does a protocol define in network communication?

(a) Hardware specifications (b) Format and order of messages (c) Host locations (d) Data encryption methods

12. Suppose two hosts are communicating from the "Distance 0". What is the best coverage area that can satisfy the connection?

(a) WAN (b) PAN (c) LAN (d) MAN (e) None

13. Rules include an identified sender and receiver, an agreed-upon method of communicating, and messages or data.

(a) True (b) False

14. The ..... in a PC converts bits into electrical signals for transmission over the network.

(a) encoder (b) decoder (c) NIC (d) medium

15. Network Core typically consists of intermediary devices like routers and servers.

(a) True (b) False

16. Which network scale typically spans over large distances, even across countries?

(a) SAN (b) PAN (c) LAN (d) MAN (e) WAN

17. What is another name for the TCP/IP reference model?

(a) OSI reference model (b) 6-layer model (c) 5-layer model (d) Both B and C

18. Which reference model is specifically used for the course?

(a) OSI reference model (b) TCP/IP 4-layer model (c) TCP/IP 5-layer model (d) Protocol 4-layer model

19. Which OSI layer is responsible for end-to-end communication between the two devices?

(a) Layer 2 (b) Layer 3 (c) Layer 4 (d) Layer 5 (e) None

20. What is layer 1 in the TCP/IP 4-layer model?

(a) Physical (b) data link (c) Internet (d) Network Access (e) Application

21. What is the PDU of the upper layer of layer 4 in the protocol model?

(a) Frame (b) Segment (c) Bits (d) Data (e) Datagram

22. What is the PDU of the lower layer of layer 2 in the OSI model?

(a) Frame (b) Segment (c) Bits (d) Data (e) Datagram

23. What is the header of the lower layer of layer 5 in the protocol model?

(a) IP address (b) Segment (c) Port number (d) MAC address (e) 11101

24. What is the header of the lower layer of layer 3 in the OSI model?
(a) IP address (b) Segment (c) Port number (d) MAC address (e) 10101

25. What happens to the PDU as it moves from Layer 4 to Layer 2?
(a) Segment → Message → Frame (b) Packet → Frame → Bit (c) Segment → Packet→ Frame (d) Frame → Packet → Segment

26. During the encapsulation, the PDU of the network layer contains the port numbers.
(a) True (b) False

27. What frequencies are used for the upstream channel in DSL?
(a) 0 to 4 kHz (b) 4 kHz to 50 kHz (c) 50 kHz to 1 MHz (d) 1 MHz to 2 MHz (e) None

28. In DSL, what component separates the data and telephone signals on the telco side?
(a) Router (b) DSLAM (c) Splitter (d) Modem (e) Telco

29. Which switching method is used by the traditional telephone system?
(a) Packet switching (b) Cell switching (c) FDM (d) Virtual switching (e) Circuit switching

30. ..... move packets from router's input to appropriate router output.
(a) Routing (b) Header (c) Forwarding (d) Encapsulation (e) Telco

31. The local forwarding table include the ..... and the output link.
(a) Port number (b) Header (c) MAC address (d) IP address (e) None

32. In store-and-forward, the full packet must arrive at a router before it can be transmitted on the next link.
(a) True (b) False

33. If arrival rate to input link exceeds transmission rate of output link for a period of time.
(a) Packets will queue (b) Packets will be dropped (c) A and B (d) A or B (e) None

34. TDM divides time into frames of fixed duration, with each frame divided into a fixed number of .....
(a) Time bands (b) Time slots (c) Frequency slots (d) Frequency bands (e) None

35. Which multiplexing technique dedicates a frequency band to each connection for the duration of the connection?
(a) SDM (b) TDM (c) FDM (d) CDM (e) None

36. Which OSI layer is most associated with the concept of circuit switching?
(a) Physical (b) data link (c) Internet (d) Network Access (e) Application

37. ..... is the difference between the highest and lowest frequencies available for network signals.
a) Frequency Spectrum (b) Time slots (c) Frequency slots (d) Frequency bands (e) None

38. ..... refers to the maximum amount of data that can be transmitted over a network.
(a) Spectrum (b) Frequency (c) Throughput (d) Bandwidth (e) None

39. ..... refers to the actual amount of data that can be transmitted over a network.
(a) Spectrum (b) Frequency (c) Throughput (d) Bandwidth (e) None

40. In circuit switching, the call takes only one circuit in each .....
(a) Link (b) Channel (c) Packet (d) A or B (e) TDM path